

RSAop - Exploration of Production Model Extensions

2026-04-20

Where we start: the paper-reported model

A 2×2 comparison: speaker architecture $\times$ semantics regime.

- Architecture — *incremental*: utterances are built word-by-word, updating the literal listener after each word; *global*: entire utterances are scored at once.
- Semantics — *recursive*: adjective thresholds adapt to the listener's running posterior at each step; *static*: thresholds fixed from the uniform prior.

Beyond the 2×2 , the model has a single rationality α (shared across dimensions), an LM prior weighted by $\beta = \exp(\log \beta)$, and hierarchical per-participant offsets δ_p on α . Length bias $\gamma = 0$ and lapse $\varepsilon = 0$ are fixed.

2×2 result:

Model	ELPD LOO	Δ ELPD	dSE	Δ /dSE	p
Inc. context-updating	-6985	0	—	—	—
Inc. context-fixed	-7001	-15.6	1.5	-10.4	< .001
Global context-updating	-7728	-742.8	45.0	-16.5	< .001
Global context-fixed	-7739	-753.2	44.8	-16.8	< .001

- Architecture effect is large: incremental $\gg$ global (Δ /dSE ≈ 16.5).
- Within incremental, context-updating helps (Δ /dSE = 10.4); within global, the two regimes are effectively tied (Δ ELPD ≈ 10).

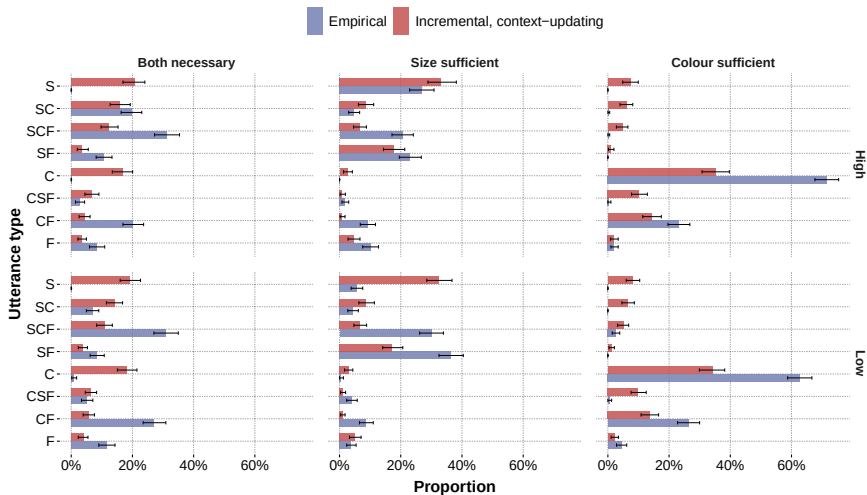
The data fit is nevertheless not optimal

- $R^2 = 0.41$ across the 90 (condition $\times$ utterance) cells.
- Neither lapse nor length bias is in the reported model: γ and ε are hardcoded to 0, so all mass comes from the RSA softmax.

Seven meaningful residuals that worth diagnosing and fixing:

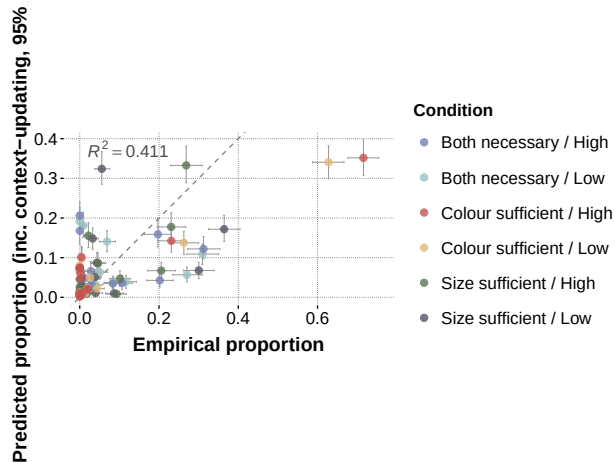
Condition	Sharpness	Utterance	Emp	Model	Δ
colour-sufficient	high	C	.716	.352	-.364
colour-sufficient	low	C	.628	.341	-.287
size-sufficient	low	D	.056	.324	+.268
size-sufficient	low	DCF	.300	.068	-.233
both-necessary	low	CF	.270	.057	-.213
both-necessary	high	D	.000	.206	+.206
both-necessary	low	DCF	.310	.109	-.201

Paper-reported model: posterior predictive check



Empirical (blue) vs predicted (red) utterance proportions by condition $\times$ sharpness. Model mass is spread too evenly: colour-sufficient C is under-predicted and both-necessary C over-predicted.

Paper-reported model: predicted vs observed



Predicted vs observed across all 90 condition × utterance cells; $R^2 = .41$.

Diagnosis

The seven residuals decompose along two largely independent axes.

1. Bare C is under-predicted when colour is sufficient

- high sharpness: $.716 \rightarrow .352$ ($\Delta = -.36$); low: $.628 \rightarrow .341$ ($\Delta = -.29$).
- A single α cannot concentrate enough mass on C in this context; the model spreads probability across C-initial compounds.
- Points to a condition-gated colour boost (λ_C in v5a).

2. Length structure: bare D over-predicted, multi-adjective utterances under-predicted

- bare D: size-sufficient / low $.056 \rightarrow .324$ ($+.27$); both-necessary / high $.000 \rightarrow .206$ ($+.21$).
- CF: both-necessary / low $.270 \rightarrow .057$ ($-.21$).
- DCF: size-sufficient / low $.300 \rightarrow .068$ ($-.23$); both-necessary / low $.310 \rightarrow .109$ ($-.20$).
- With $\gamma = 0$ in the reported model there is no utility boost for longer utterances, so the model defaults to the shortest informative form. Over-specification is also larger under blur and asymmetric across conditions, so no monotone linear γ fits all three.
- Points to an explicit length bias ($\gamma, \gamma_1, \gamma_2$) plus condition- ($\delta\gamma$) and sharpness-dependent (η) offsets.

These two diagnostics motivate the mechanism track below.

Why these diagnostics are hard to fix jointly

A single α cannot concentrate and spread at once.

- Axis 1 wants near-degenerate mass on bare C; axis 2 wants mass spread across multi-adjective forms.
- Raising α concentrates everywhere, lowering it spreads everywhere — a single α lands at a compromise.

A single linear γ cannot be selectively positive.

- Axis 2 wants a length boost in both-necessary and blurred size-sufficient — but not in colour-sufficient, and smaller under high sharpness.
- Any γ large enough to recover DCF in size-sufficient also inflates multi-adjective forms in colour-sufficient. v5b confirms: no condition gating, no gain (+6 ELPD).

The two axes are coupled through α .

- Sharpening bare C in colour-sufficient also sharpens bare D in size-sufficient.
- Per-dim α (ext-v1) helps but is a property of the speaker, not of the context — it cannot say “concentrate on C *only when* colour is sufficient”. That is what λ_C , $\delta\gamma$, and η are designed to supply.

Model Extensions Explored so far (in time order)

erdc, zrdc, brdc — size $\times$ colour distinguishing, form redundant.

Three relevant `property` values inside: `first` (size sufficient), `both`, `second` (colour sufficient).

Step	Mechanism	New parameters
ext-v1	per-dim α , γ , ε	$\alpha_D, \alpha_C, \alpha_F, \gamma, \varepsilon$
v2 (rejected)	first-word intercepts	ϕ_C, ϕ_F
v3 (rejected)	step-level logit offsets (per-dim α)	$\mu_C, \mu_F + \text{per-dim } \alpha$
v4 (rejected)	step-level logit offsets (single α)	μ_C, μ_F
mix-pd (abandoned)	two-speaker mixture, per-dim α	$\alpha_{D,C,F}^{(j)}, \pi, \varepsilon$
mix-s (abandoned)	two-speaker mixture, single α	$\alpha^{(j)}, \gamma^{(j)}, \pi, \varepsilon$
v5a	condition-gated colour boost	λ_C (step-0 only)
v5b	saturating length bias	γ_1, γ_2
v5 (F1)	+ csuff length offsets	$\delta\gamma_1, \delta\gamma_2$
v5 (C1)	+ canonical-order penalty	μ_{nc}
v5 (F2)	+ sharpness-dependent length	η_1, η_2

Note: the extension track uses the code defaults $csv = 0.971$, $fsv = 0.50$; the paper uses $csv = 0.85$, $fsv = 0.70$. The ΔELPDs reported below are internally consistent within the extension baseline, not directly comparable to the paper's Set B numbers.

Stage 1: ext-v1

Per-dimension α + explicit γ + lapse ε .

	ELPD	vs baseline	R^2	L1
baseline	-7157	0	.35	3.17
ext-v1	-6387	+770	.60	3.17

Posterior: $\alpha_D = 6.92$, $\alpha_C = 1.85$, $\alpha_F = 1.89$, $\gamma = 2.31$, $\varepsilon = 0.23$.

Seven systematic misfits persist, and $\varepsilon = 0.23$ is high enough to suggest unexplained structure.

Rejected pre-v5 extensions

Before v5, three condition-independent biases were tested; all were rejected.

Variant	Mechanism	R^2	ELPD	Outcome
v2 (first-word intercepts)	ϕ_C, ϕ_F on the chosen utterance type's first adjective (S = reference)	.62	−6337	Wrong sign: $\phi_C = -6.89$ (penalty) rather than boost. The misfit is context-dependent, so a condition-independent intercept cannot capture “C up in csuff, C down in both-necessary”.
v3 (step-level μ w/ per-dim α)	logits = $\alpha_{\text{vec}} \cdot \log L_{\text{ref}} + \mu_{\text{vec}}$	—	—	Non-identifiable: $n_{\text{eff}} \approx 2$, $\hat{R}$ up to 14.7 for μ_C . α and μ trade off on the same logit scale — structural, not a sampling issue.
v4 (step-level μ w/ single α)	Dropped per-dim α ; kept μ_C, μ_F to avoid v3's identifiability problem	.54	−6449	Worse than ext-v1. Losing per-dim α hurt more than μ helped, which suggests that per-dim α captures something that μ does not.

Takeaway for v5: the misfit is context-dependent rather than a global utterance-type tilt. This motivated the condition-gated λ_C in v5a.

Alternative track: two-speaker mixture — setup

A separate attempt at accounting for the residuals: strategy heterogeneity across participants, modelled as a two-component mixture over incremental recursive speakers.

Shared structure.

- Each component $j \in \{1, 2\}$ is an incremental recursive speaker with its own rationality and length bias.
- Mixing weight $\pi \sim \text{Beta}(2, 2)$; lapse $\varepsilon \sim \text{Beta}(2, 98)$.
- Label-switching constraint: $\text{gap}_D \sim \text{HalfN}(3)$, with $\alpha_{2,D} = \max(\alpha_{1,D} - \text{gap}_D, 0)$.

Two variants.

- **mix-pd**: per-component per-dim $\alpha \rightarrow \alpha_D^{(j)}, \alpha_C^{(j)}, \alpha_F^{(j)}, \gamma^{(j)}$ per speaker (9 per-component parameters in total with the gap constraint).
- **mix-s**: per-component single $\alpha \rightarrow \alpha^{(j)}, \gamma^{(j)}$ per speaker (2 per-component parameters). Designed to reduce identifiability pressure.

Mixture variant 1: per-dim α (mix-pd)

Most informative run: full data ($N = 9586$), long warmup.

Parameter	Speaker 1	Speaker 2	Interpretation
α_D	14.82	8.57	both deliberate about size
α_C	20.85	1.24	speaker 1 super-rational on colour, speaker 2 barely
α_F	6.79	1.29	speaker 1 also deeper on form
γ	-2.29	+1.76	1 penalises length; 2 rewards length
π	0.40 $\leftrightarrow$ 0.60		$\sim 40\%$ “careful” / 60% “mention-more”
ε	0.15		

- Qualitatively the two sub-populations separate: a minimum-description speaker and a mention-more speaker.
- But $\hat{R} \in [1.4, 2.8]$ despite 0 divergences — NUTS finds both modes, chains do not mix.
- On the dc subset the components do not even separate ($\text{gap}_D \approx 0.1$).

Mixture variant 2: single α (mix-s)

Same mixture shape with only $\alpha^{(j)}$ and $\gamma^{(j)}$ per component — chosen to reduce per-component parameter count and ease identifiability.

Long-warmup run (5000 warmup).

- α_1 converges ($\hat{R} = 1.04$).
- α_2 : $\hat{R} = 2.80$.
- γ_2 does not converge.

Subset run (short).

- Component 2 degenerates toward uniform; ε inflates to absorb the resulting slack.

Reducing the per-component parameter count does not resolve the mixing problem. The remaining non-identifiability is structural, not a parameter-count issue — more likely a NUTS-on-mixture limitation.

Mixture takeaway and why the track was set aside

Qualitatively the mixture story holds.

- Two sub-populations separate cleanly in the long run on the full data.
- Speaker 1 ($\alpha_C = 20.9$, $\gamma = -2.3$): utility-reasoning, minimum-description.
- Speaker 2 ($\alpha_C = 1.2$, $\gamma = +1.76$): low- α , mention-more default.
- This matches the strategy-heterogeneity reading that ext-v1's high $\varepsilon = 0.23$ was absorbing.

Quantitatively, the mixture is not yet identified. Every variant shows $\hat{R} \geq 1.4$ on key parameters despite 0 divergences; multimodal posteriors; subset runs do not even separate the components. NUTS alone cannot traverse the mixture modes at this sample size, and this is unlikely to be fixable by tuning — would need EM or variational inference.

Relation to v5.

- v5 models a single-strategy speaker with context-dependent mechanisms (λ_C , μ_{nc} , $\delta\gamma$, η).
- The mixture models a strategy-heterogeneous speaker without those mechanisms.
- A combined v5 + mixture model would test whether the two accounts are redundant or complementary. Not yet run.

Stage 2a: λ_C alone (v5a)

Step-0 logit boost for **C** when `is_colour_sufficient = 1`.

	ELPD	Δ vs ext-v1	λ_C posterior
ext-v1	-6387	0	—
v5a	-6265	+122	3.4 (95% CI 2.8–4.0)

- λ_C is clearly positive: colour gets extra first-step pull when it suffices.
- **second/sharp C**: empirical .716, previously .403, now $\sim$.55.
- CF is still too high in **second**, because the boost also spreads to C-initial compounds.

Stage 2b: saturating γ alone (v5b)

Replace linear $\gamma \cdot k$ with $\gamma_1 \mathbb{I}[k \geq 1] + \gamma_2 \mathbb{I}[k \geq 2]$.

	ELPD	Δ vs ext-v1
ext-v1	-6387	0
v5b	-6381	+6

Essentially no improvement without λ_C .

The two mechanisms interact: γ -saturation only starts to matter once C-initial mass is correctly apportioned.

Stage 3: F1 — condition-dependent length ($v5 + \delta\gamma$)

Let the length bias **shrink** on colour-sufficient trials:

$$\gamma_{1,\text{eff}} = \gamma_1 + \delta\gamma_1 \cdot \text{is_csuff}.$$

	ELPD	Δ vs ext-v1
ext-v1	-6387	0
v5a	-6265	+122
v5 (F1)	-5933	+454

Posteriors: $\lambda_C = 5.0$, $\gamma_1 = 2.56$, $\gamma_2 = 0.92$, $\delta\gamma_1 = -1.89$, $\delta\gamma_2 = -3.51$, $\varepsilon = 0.16$.

Effective γ on csuff trials: 0.67, -2.59 — 3-word heavily penalised, short utterances favoured.

$R^2 = .67$, $L1 = 2.70$. Remaining residuals: DC and DFC over-predicted, DCF under-predicted.

Stage 4: C1 — canonical-ordering penalty

Add $\mu_{\text{nc}} \cdot \mathbb{I}[\text{utterance violates } D < C < F]$.

Non-canonical set: CD, CDF, CFD, DFC, FD, FDC, FC, FCD.

	ELPD	Δ vs ext-v1
v5 (F1)	-5933	+454
v5 (C1)	-5323	+1063

$\mu_{\text{nc}} = -5.07$ — a strong canonical-English preference.

ε drops to 0.10.

$R^2 = .91$, $L1 = 1.34$.

Most F1 residuals are resolved: canonical-order cells (DCF, DFC, DC, DF) and bare C now match empirical within $\sim .05$.

One side-effect: on *first/sharp*, bare D is under-predicted (.082 vs. empirical .269) and DF over-predicted.

Stage 5: F2 — sharpness-dependent length (final)

Add η_1, η_2 offsets active on `is_sharp` trials.

	ELPD	Δ vs ext-v1	R^2
v5 (C1)	-5323	+1063	.91
v5 (F2)	-5294	+1093	.94

Posteriors: $\eta_1 = -0.61$, $\eta_2 = -0.34$ (both credibly negative).

Effective γ on **sharp** trials: 1.71, 1.24 vs. on **blurred**: 2.32, 1.58.

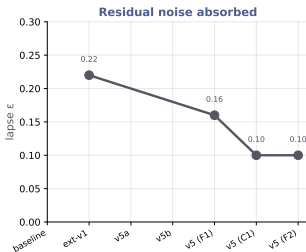
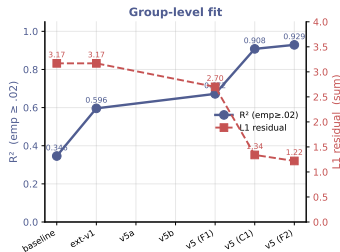
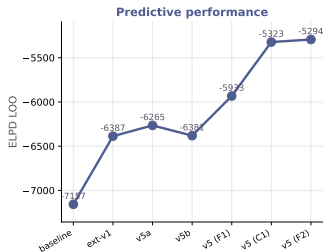
ε stays at 0.10; 0 divergences; all $\hat{R} \leq 1.02$.

One residual persists: `first/sharp` D is still under-predicted ($.269 \rightarrow .102$).

Relation to the strategy heterogeneity: some participants describe minimally, others overinformatively.

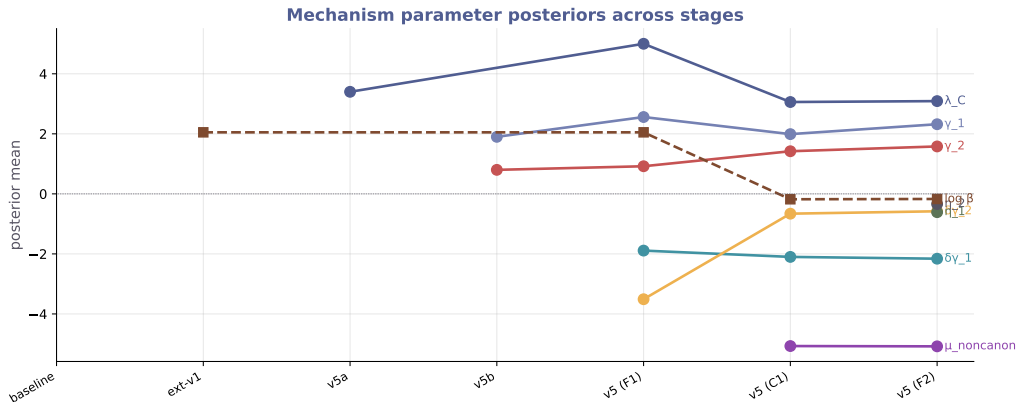
Fit evolution across stages

Extension v5 — Fit evolution across mechanism additions (dc subset, N = 3196)



Each new mechanism shifts LOO and R^2 upward and ϵ downward. The exception is v5b (saturating γ alone), which is flat against ext-v1.

Mechanism posteriors across stages



Each mechanism parameter enters with a consistent posterior sign and settles by F2. λ_C and μ_{nc} are strong, η is modest, $\delta\gamma_1$ is clearly negative on csuff, and $\delta\gamma_2$ shrinks once μ_{nc} absorbs canonical-order variance. $\log \beta$ (dashed) is pre-existing and drops sharply from ~ 2 to -0.18 once μ_{nc} is added, indicating that canonical ordering absorbs much of what the LM prior was previously encoding.

Ablation: is the LM prior still needed under F2?

Test: fit v5 (F2) with $\beta = 0$, effectively removing the LM prior. Every other mechanism is held fixed.

Model	ELPD	Δ vs F2	R^2	L1
v5 (F2)	-5294	0	.928	1.22
v5_no_lm (F3)	-5314	-20.3	.915	1.30

Posterior compensation when the LM prior is removed:

- λ_C : 3.09 $\rightarrow$ 3.33
- γ_2 : 1.58 $\rightarrow$ 1.32
- $\delta\gamma_1$: -2.16 $\rightarrow$ -2.39
- $\mu_{nc}, \gamma_1, \eta, \varepsilon$: essentially unchanged.

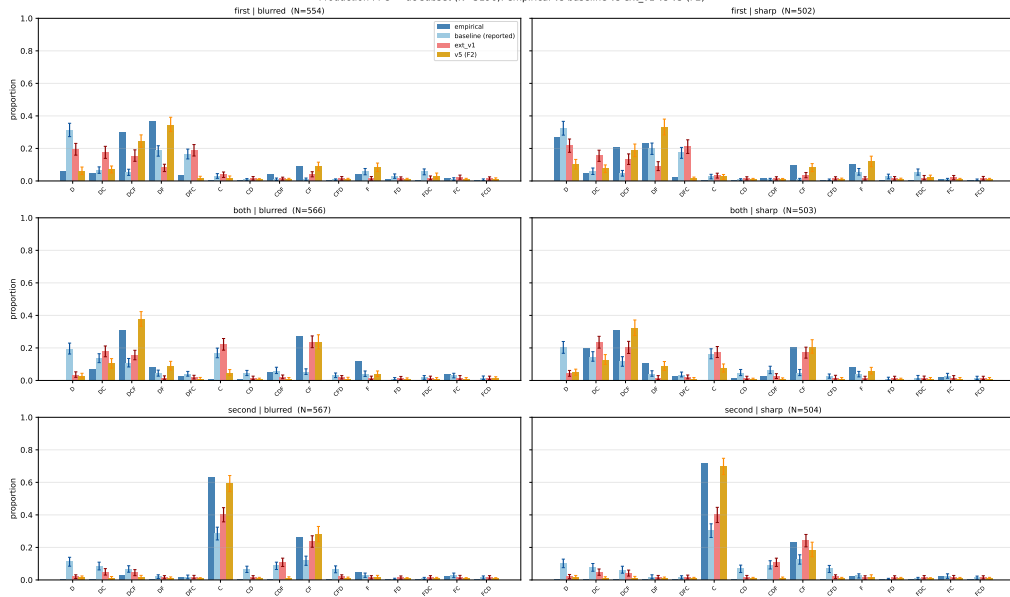
LM prior contribution goes most likely to frequency structure beyond canonical ordering and length (e.g., relative frequency among same-shape canonical utterances)

Final v5 (F2) parameters — full record

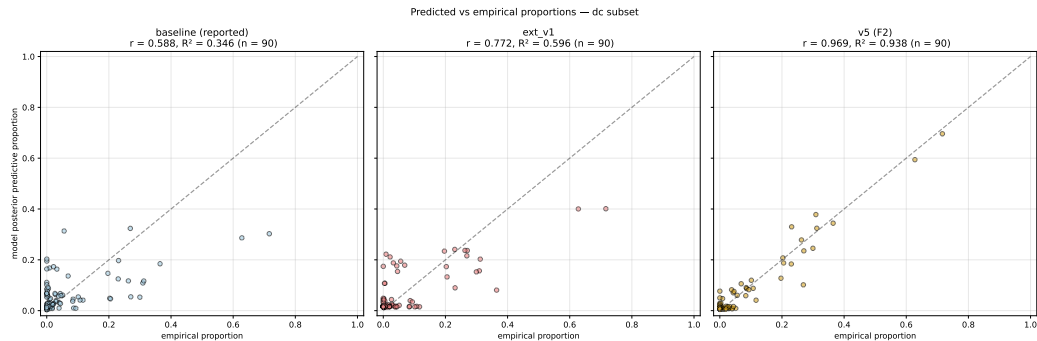
Parameter	Prior	Posterior mean (95% CI)	$\hat{R}$
α_D	HalfN(5)	(hier. w/ δ_p)	≤ 1.01
α_C	HalfN(5)	(hier.)	≤ 1.01
α_F	HalfN(5)	(hier.)	≤ 1.01
λ_C	N(0,1)	+3.09 (2.67, 3.55)	1.00
γ_1	N(0,1)	+2.32 (2.16, 2.49)	1.00
γ_2	N(0,1)	+1.58 (1.43, 1.74)	1.01
$\delta\gamma_1$	N(0,1)	-2.16 (-2.33, -1.97)	1.00
$\delta\gamma_2$	N(0,1)	-0.58 (-1.21, +0.04)	1.00
η_1	N(0,1)	-0.61 (-0.77, -0.43)	1.00
η_2	N(0,1)	-0.34 (-0.51, -0.15)	1.00
μ_{nc}	N(0,1)	-5.08 (-5.61, -4.56)	1.01
$\log \beta$	N(0,0.5)	-0.17	1.00
ε	Beta(1,50)	0.10 (0.09, 0.12)	1.00
τ	HalfN(0.2)	0.62 (0.51, 0.73)	1.00

PPC — v5 (F2) on the dc subset

Production PPC — dc subset (N=3196): empirical vs baseline vs ext_v1 vs v5 (F2)



Predicted vs observed — v5 (F2) on the dc subset



Predicted vs observed proportions across the 30 (condition $\times$ sharpness $\times$ utterance) cells on the dc subset. R^2 :
baseline .35 $\rightarrow$ ext-v1 .60 $\rightarrow$ **v5 (F2) .94.**

Generalisation to the full data — proof of concept

We refit the *same* F2 structure (same mechanisms, same priors, no re-tuning) on the full $N = 9586$ dataset spanning all 9 conditions.

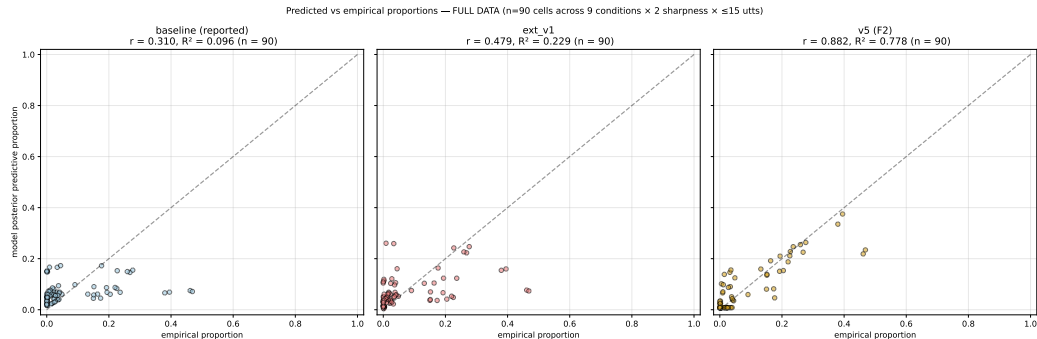
Model	ELPD LOO	R_{90}^2	R_{270}^2	L1
baseline	-22703	.10	—	6.46
ext-v1	-20831	.23	—	5.46
v5 (F2)	-15969	.78	.73	2.68

R_{90}^2 : 90 (condition-family $\times$ utterance) cells marginalised across sharpness. R_{270}^2 : full 9 conditions $\times$ 2 sharpness $\times$ 15 utterances.

- +4862 ELPD over ext-v1 on full data (vs. +1093 on the dc subset) — the gap widens roughly 4.5 $\times$.
- μ_{nc} posterior: -5.08 (dc) $\rightarrow$ -8.30 (full) — the canonical-ordering penalty strengthens over the wider condition space.

Treat this as a proof of concept rather than a formal out-of-sample test: the mechanisms were designed on the dc subset and refit here, and no posterior checks beyond overall fit metrics have been done yet.

Predicted vs empirical — full-data fit ($n = 90$ cells marginalised)



All 90 condition-family $\times$ utterance cells. R^2 : baseline .10 $\rightarrow$ ext-v1 .23 $\rightarrow$ **v5 (F2) .78** on full-data fit.

Re-running the 2×2 under F2

With the F2 mechanisms in place, we re-ran the 2×2 (speaker architecture $\times$ semantics regime) on the dc subset.

Two parameterisations to compare in the main text:

- 1 Paper-reported: minimal, theory-driven.
- 2 F2-full globals: every parameter sampled freely in every cell — the comparison that holds expressive capacity equal.

2×2 comparison with F2

	recursive	static	$\Delta(\text{rec} - \text{sta})$
incremental	−5294	−5297	+2.80 ($\Delta/\text{dSE} = 2.22$)
global (full)	−5357	−5365	+8.25 ($\Delta/\text{dSE} = 3.82$)
$\Delta(\text{inc} - \text{glb})$	+63	+69	
Δ/dSE	3.18	3.48	

Key changes relative to the paper-reported numbers:

- Architecture effect Δ/dSE : $11 \rightarrow 3.2$ (roughly a third of the paper's).
- Regime-within-inc Δ/dSE : $10.28 \rightarrow 2.22$ (roughly a fifth of the paper's).
- Regime-within-glb: $\approx 0 \rightarrow 3.82$ (a modest effect now appears).
- Diff-in-diff: $+21 \rightarrow -5.45$ — magnitude shrinks and sign reverses (recursive now helps global slightly more than incremental).

Full record — side-by-side

	Paper-reported	F2-full
Sampled parameters	$\alpha, \log \beta, \tau, \delta_p$	$+ \alpha_D \alpha_C \alpha_F$ (hier), $\lambda_C, \gamma_{1,2}, \delta\gamma_{1,2}, \eta_{1,2}, \mu_{nc}, \varepsilon$
Fixed	$\gamma=0, \varepsilon=0$	— (all of above sampled)
R^2 (all cells)	.35	.94
L1 residual	3.17	1.22
Lapse ε	0 (fixed)	0.10 (sampled)
Arch effect Δ/dSE	± 11	± 3.2
Regime inc Δ/dSE	10.28	2.22
Regime glb Δ/dSE	≈ 0	3.82
Interaction (diff-in-diff)	+21	−5.45
Rec helps. . .	incremental	slightly global

F2-full adds 9 sampled mechanism parameters on top of the paper-reported set, plus per-dim hierarchical α and a sampled lapse. The F2-fixed parameterisation (globals frozen at inc-recursive posterior means) is moved to the appendix.

The tradeoff between the two tracks

Theory track (paper):

- Minimal, motivated mechanisms.
- Clear 2×2 interaction.
- Group-level fit is poor ($R^2 = .35$).
- Seven large misfits in §5.3.
- Lapse absorbs unexplained structure.

Mechanism track (F2):

- Richer; each term is literature-motivated.
- Much better fit ($R^2 = .94$).
- All seven misfits resolved.
- Lapse halved ($0.22 \rightarrow 0.10$).
- Architecture signal shrinks to $\Delta/\text{dSE} = 3$.
- Regime-within-inc weakens to marginal.

The mechanisms we added to reduce residuals also absorb variance that the paper's minimal parameterisation had attributed to architecture $\times$ regime. The interaction survives, but in reduced form.

Two readings of the tradeoff

Reading I — the mechanisms are rivals to the architectural account.

The paper's interaction was partly architectural sparsity fitting error structure. Once the error is absorbed by explicit mechanisms (colour salience, canonical ordering, context-dependent length), the residual architectural signal is modest, and the main claim has to be weakened accordingly.

Reading II — the mechanisms are what the architecture was *supposed* to capture but fails to.

λ_C , μ_{nC} , $\delta\gamma$, and η are all condition-dependent context effects — exactly the kind of structure that a context-updating incremental speaker was introduced to model. On this reading the architecture fails in practice for one of two reasons: (i) the context signal is too subtle for threshold updates to pick up (the architecture exists but is not sensitive enough), or (ii) human behaviour is qualitatively different from what threshold updates predict (e.g., a first-step colour boost or a canonical-order preference that no threshold update can reproduce). Either way, making these mechanisms explicit is how the model starts tracking the context the architecture was meant to track.

Both readings are compatible with the numbers; the choice is mainly a question of how we want to frame the paper.

Summary

- The paper-reported model is theory-driven and yields an interpretable 2×2 , but the fit is poor ($R^2 = .35$).
- The residuals trace to three largely independent phenomena: colour salience, canonical ordering, and condition-dependent length bias.
- Adding all three (v5 F2) lifts the fit to $R^2 = .94$ and brings the lapse down to .10, with every new parameter credibly non-zero and 0 divergences.
- Re-running the 2×2 under F2: the architecture effect shrinks by roughly a factor of three, and the regime effect shrinks and partially flips direction.

Thanks

Questions and discussion.

Appendix: paper-reported model parametrization

Sampled parameters.

Parameter	Role	Prior
α	rationality (shared across D, C, F)	HalfNormal(5.0)
$\log \beta$	LM prior weight, $\beta = \exp(\log \beta)$	Normal(0, 0.5)
τ	participant random-effect scale on α	HalfNormal(0.2)
δ_p	per-participant offset on α	Normal(0, τ)

Fixed (not sampled) parameters.

Parameter	Role	Value
γ	length bias	0
ε	lapse	0
k	size threshold	0.5
w_f	form weight	1.0
ν_{colour}	colour semantic value	0.85
ν_{form}	form semantic value (production only)	0.70

MCMC settings (NumPyro NUTS).

- 4 chains, 5,000 warmup and 2,000 post-warmup samples per chain.
- Target acceptance probability = 0.85, maximum tree depth = 7.
- Convergence confirmed via split- $\hat{R} < 1.01$ and $n_{\text{eff}} > 400$ across all parameters.

Appendix: why not free the semantic parameters?

An alternative to the mechanism track is to sample k , ν_{colour} , ν_{form} , and w_f instead of fixing them. These parameters directly control the utility imbalance behind the residuals (currently colour ≈ 1.65 nats, size ≈ 0.06 nats, form ≈ 0 nats).

Identifiability — profile-posterior analysis (Feb 2026).

- k is not identifiable: the profile posterior is flat, because the size manipulation is not contrastive enough to pin k down.
- ν_{colour} and ν_{form} are each unimodal individually, but the joint posterior is bimodal and NUTS chains do not mix.
- w_f : sharpening (lower w_f) consistently hurts in grid search.

Grid search over fixed values (March 2026).

- Lowering ν_{colour} from 0.971 to 0.85 is the only change that helps: +137 ELPD on the reported model (3-cond subset), +245 on full data, with p_{loo} dropping from 93 to 67.
- Form and size-sharpness variations are negligible or harmful.

The semantic parameters are therefore not jointly identifiable, so freeing them is not a viable substitute for the mechanism track. Fixed-value search points in the same qualitative direction — colour's effective discriminability should be lower than its literal reading — and v5 adds identifiable handles on top of this fixed-representation baseline.

Appendix: 2×2 with F2 globals fixed at inc-recursive means

Sanity-check parameterisation: the global cells use $\gamma, \delta\gamma, \eta, \mu_{nc}$ fixed at the incremental-recursive posterior means, rather than sampled freely.

	recursive	static	$\Delta(\text{rec-sta})$
incremental	-5294	-5297	+2.80 ($\Delta/\text{dSE} = 2.22$, marginal)
global (fixed)	-5600	-5619	+19.15 ($\Delta/\text{dSE} = 6.99$)
$\Delta(\text{inc-glb})$	+306	+323	
Δ/dSE	10.51	11.03	

- The architecture effect is still large ($\Delta/\text{dSE} \approx 10.5$).
- The interaction has flipped: recursive now helps global, not incremental.
- This is not a like-for-like test — the global models are constrained to inc-recursive parameter values. F2-full globals is the fair comparison.

Appendix: full LOO & fit record — all stages

Stage	ELPD LOO	Δ vs baseline	Δ vs ext-v1	R^2	L1	ε
baseline	−7157	0	—	.35	3.17	—
ext-v1	−6387	+770	0	.60	3.17	.23
v2 (first-word intercepts) — rejected	−6337	+820	+50	.62	—	—
v3 (step-level μ , per-dim α) — rejected	non-identifiable ($\hat{R} \leq 14.7$)					
v4 (step-level μ , single α) — rejected	−6449	+708	−62	.54	—	—
v5a (λ_C only)	−6265	+892	+122	—	—	—
v5b (sat. γ only)	−6381	+776	+6	—	—	—
v5 (F1: $\lambda_C + \delta\gamma$)	−5933	+1224	+454	.67	2.70	.16
v5 (C1: + μ_{nc})	−5323	+1834	+1063	.91	1.34	.10
v5 (F2: + η)	−5294	+1863	+1093	.94	1.22	.10
v5_no_lm (F3: F2 with $\beta = 0$)	−5314	+1843	+1073	.92	1.30	.10
<i>Full data generalization check (N = 9586, all 9 conditions):</i>						
baseline (full data)	−22703	0	—	.10	6.46	—
ext-v1 (full data)	−20831	+1872	0	.23	5.46	.05
v5 (F2, full data)	−15969	+6734	+4862	.78	2.68	.12

- Columns: LOO, R^2 over cells with empirical $\geq .02$, L1 (sum of $|\Delta|$ over 37 cells), and posterior mean of ε .
- Baseline has no lapse and no length bias: γ and ε are hardcoded to 0 in `likelihood_function_reported_hier`, not sampled. Entries marked “—” are structurally absent, not unreported.
- v2–v4: rejected exploratory variants

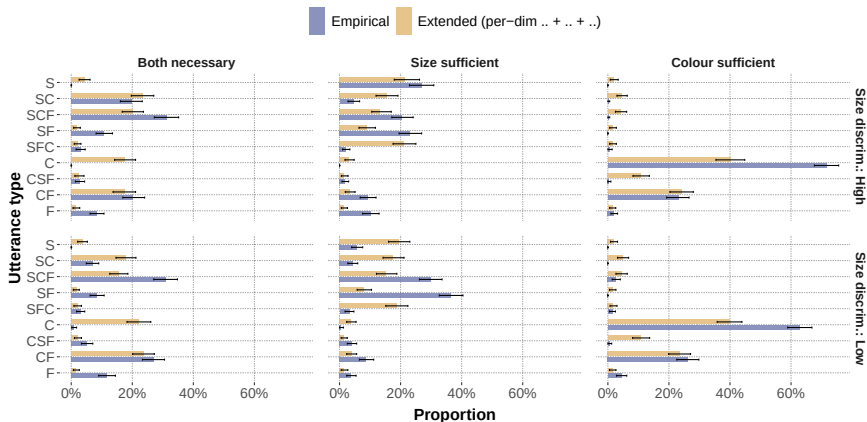
Appendix: full parameter record — posterior means across stages

Parameter	baseline	ext-v1	v2	v4	v5a	v5b	v5 (F1)	v5 (C1)	v5 (F2)
α_D	single α	6.92	~ 6.9	single α	~ 6.9	~ 6.9	hier.	hier.	hier.
α_C		1.85	~ 1.8		~ 1.8	~ 1.9	hier.	hier.	hier.
α_F		1.89	~ 1.9		~ 1.9	~ 1.9	hier.	hier.	hier.
ϕ_C (first-word)	—	—	−6.89	—	—	—	—	—	—
ϕ_F (first-word)	—	—	−5.90	—	—	—	—	—	—
μ_C (step-level)	—	—	—	fitted	—	—	—	—	—
μ_F (step-level)	—	—	—	fitted	—	—	—	—	—
λ_C	—	—	—	—	+3.4	—	+5.0	+3.06	+3.09
γ (linear)	—	2.31	~ 2	~ 2	—	—	—	—	—
γ_1	—	—	—	—	—	+1.9	+2.56	+1.99	+2.32
γ_2	—	—	—	—	—	+0.8	+0.92	+1.42	+1.58
$\delta\gamma_1$	—	—	—	—	—	—	−1.89	−2.10	−2.16
$\delta\gamma_2$	—	—	—	—	—	—	−3.51	−0.66	−0.58
η_1	—	—	—	—	—	—	—	—	−0.61
η_2	—	—	—	—	—	—	—	—	−0.34
μ_{nc}	—	—	—	—	—	—	—	−5.07	−5.08
$\log \beta$	~ 2	2.05	~ 2	~ 2	~ 2	~ 2	2.05	−0.18	−0.17
ε	—	0.23	$\sim .2$	$\sim .2$	—	—	0.16	0.10	0.10
τ	—	0.92	—	—	—	—	0.94	0.60	0.62

v3 omitted (non-identifiable).

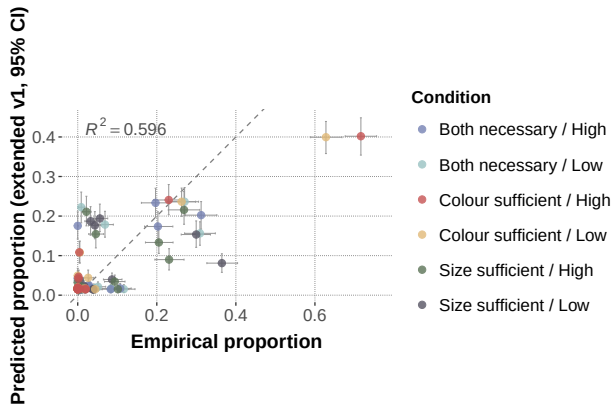
■ “hier.” = α sampled hierarchically with per-participant δ_p ; population means close to ext-v1’s values.

Appendix: ext-v1 posterior predictive check



Empirical (blue) vs ext-v1 predicted (gold) utterance proportions by condition $\times$ sharpness. Per-dim α and explicit γ lift bare C in colour-sufficient and recover some mass on CF and DCF; residual misfits remain.

Appendix: ext-v1 predicted vs observed



Predicted vs observed across all 90 condition $\times$ utterance cells. R^2 improves from .41 (paper-reported) to .60.

Appendix: ext-v1 — what improved and what persists

What the extension closes.

- Per-dim α ($\alpha_C = 1.85$ vs. the reported single $\alpha \approx 1.0$) partially lifts bare C in colour-sufficient.
- Explicit $\gamma = 2.31$ gives multi-adjective utterances (CF, DCF) a utility boost, closing most of the bare-D over-prediction seen in the paper-reported model.
- Lapse $\varepsilon = 0.23$ absorbs the remainder.

Residuals that persist (seven largest).

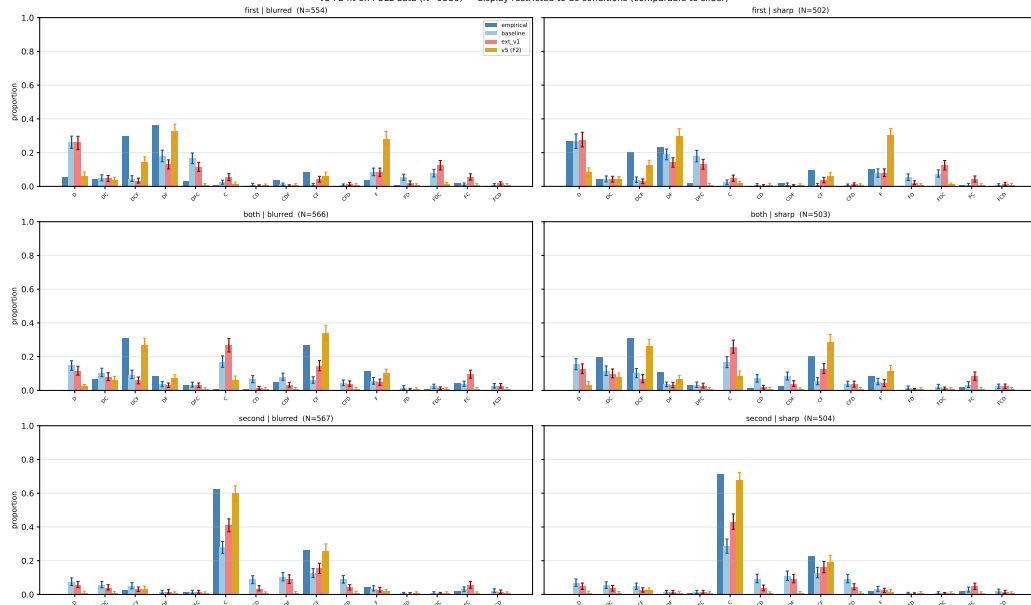
Condition	Sharpness	Utterance	Emp	ext-v1	Δ
colour-sufficient	high	C	.716	.402	-.314
colour-sufficient	low	C	.628	.399	-.229
both-necessary	low	C	.009	.222	+.213
both-necessary	high	C	.000	.175	+.175
size-sufficient	low	DF	.365	.081	-.284
size-sufficient	high	DFC	.022	.208	+.186
size-sufficient	low	DFC	.032	.187	+.155

Two diagnostic axes remain:

- Bare C is still context-dependent: under-predicted in colour-sufficient and over-predicted in both-necessary. A condition-independent α_C cannot flip sign across contexts.
- Length residuals saturate: DF under-predicted in blurred size-sufficient, DFC over-predicted in size-sufficient — no monotone scalar α fits both

Appendix: full-data F2 fit displayed on dc cells

v5 F2 fit on FULL data (N=9586) — display restricted to dc conditions (comparable to slider)



Appendix: full-data F2 fit on all 9 conditions

